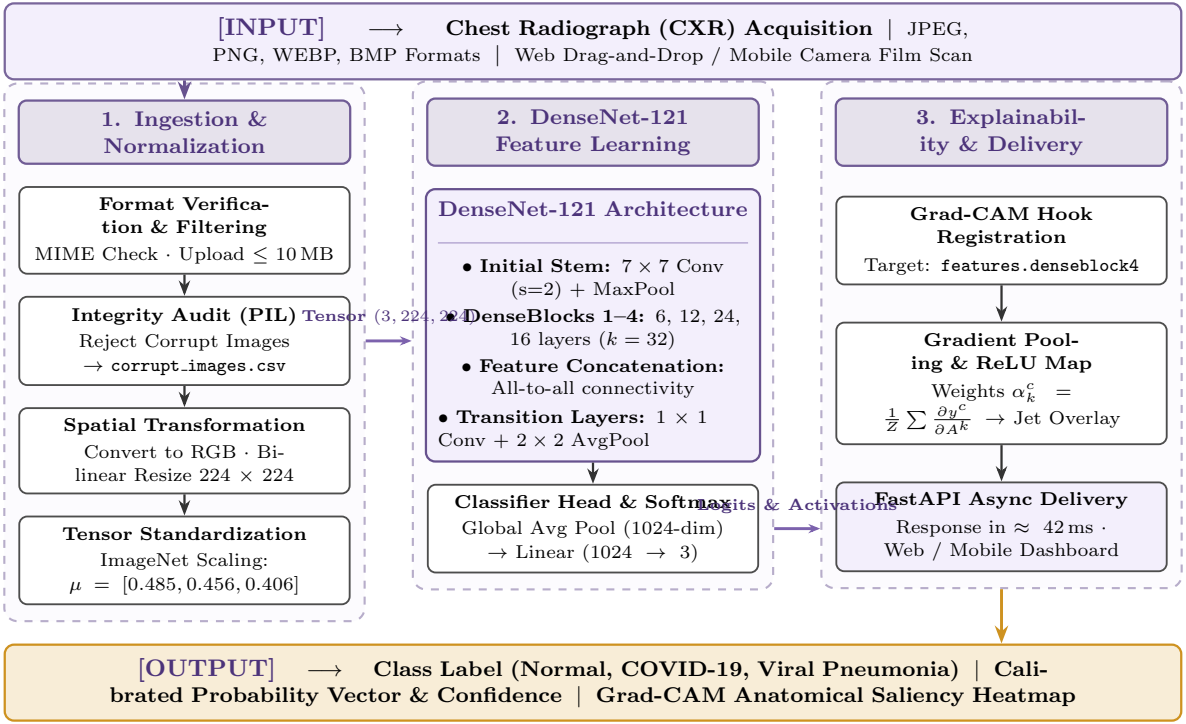

[INPUT] ⟶ Chest Radiograph (CXR) Acquisition | JPEG, PNG, WEBP, BMP Formats | Web Drag-and-Drop / Mobile Camera Film Scan
1. Ingestion & Normalization
Format Verification & Filtering
MIME Check · Upload ≤ 10 MB
Integrity Audit (PIL)
Reject Corrupt Images
→ corrupt_images.csv
Spatial Transformation
Convert to RGB · Bilinear Resize 224 × 224
Tensor Standardization
ImageNet Scaling:
μ = [0.485, 0.456, 0.406]
Tensor (3, 224, 224)
2. DenseNet-121 Feature Learning
DenseNet-121 Architecture
• Initial Stem: 7 × 7 Conv (s=2) + MaxPool
• DenseBlocks 1–4: 6, 12, 24, 16 layers (k = 32)
• Feature Concatenation: All-to-all connectivity
• Transition Layers: 1 × 1 Conv + 2 × 2 AvgPool
Classifier Head & Softmax
Global Avg Pool (1024-dim)
→ Linear (1024 → 3)
Logits & Activations
3. Explainability & Delivery
Grad-CAM Hook Registration
Target: features.denseblock4
Gradient Pooling & ReLU Map
Weights $\alpha_k^c = \frac{1}{Z}\sum \frac{\partial y^c}{\partial A^k}$ → Jet Overlay
FastAPI Async Delivery
Response in ≈ 42 ms · Web / Mobile Dashboard
[OUTPUT] ⟶ Class Label (Normal, COVID-19, Viral Pneumonia) | Calibrated Probability Vector & Confidence | Grad-CAM Anatomical Saliency Heatmap